TSGI-SM: Text-Subject-Guided Image Inpainting via Sketch Mediation

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Abstract. Recent advances in multimodal condition-guided image inpainting, particularly text-subject-guided methods, offer enhanced control dimensions over unimodal approaches. However, achieving effective multimodal alignment presents a significant challenge. Existing methods often suffer from Faithfulness Hallucination—systematic deviations between the generated content and the multiple input modalities (prompts or subject images), particularly when performing structural edits. Furthermore, mainstream solutions rely on costly large-scale training and per-subject fine-tuning, leading to high costs. To address these challenges, we propose a sketch-mediated framework that leverages a sketch as a powerful intermediary for cross-modal structured representation. First, a text-guided sketch generation module translates texts into explicit visual structural representations. Second, a training-free inpainting module jointly utilizes the sketches and the subject image to guide the image inpainting. The subject image provides detailed visual information, while the sketch clarifies the structural goals and layout for inpainting, enabling precise refinement. Our zero-shot method lowers training costs and delivers a 80× faster response than few-shot methods. Experiments demonstrate that TSGI-SM effectively addresses the challenge of multimodal alignment, outperforming SOTA in subject identity preservation (+15.8%) and text-guided structural editing accuracy (+16.4%).[4]

Keywords: Multimodality · Image inpainting · Diffusion model.

1 Introduction

Traditional image inpainting techniques use the image's intrinsic semantics to fill masked regions. In contrast, conditional image inpainting restores images based on masks and specific conditions, like text [23, 20, 1, 2] or images [27, 22, 4, 14, 17], enabling customized edits with auxiliary information. The goal is to achieve

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[4] Our code is available at https://anonymous.4open.science/r/TSGI-SM-B68F/

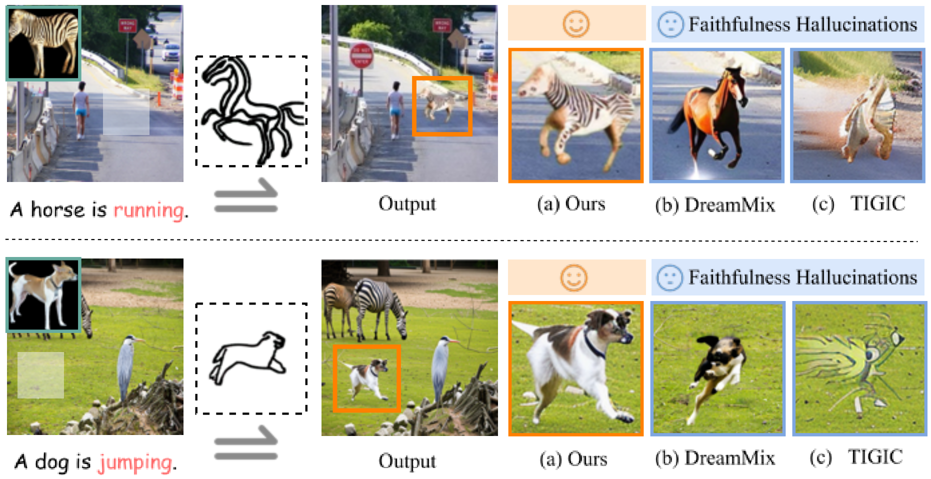

Fig. 1. TSGI-SM examples vs. Faithfulness Hallucinations in other methods. (b) DreamMix [28] and (c) TIGIC [13] are strong baselines but suffer from modality misalignment, leading to Faithfulness Hallucination, where generated results mismatch text semantics or subject identity. In contrast, (a) our TSGI-SM introduces a sketch-mediated mechanism that aligns textual and visual modalities, enabling faithful and identity-consistent edits for precise text-subject-guided image inpainting.

precise control over the reconstruction of missing regions, useful for applications like scarce data generation and virtual try-on.

However, unimodal conditional image inpainting is limited in editing precision for inserted objects. To address this, recent studies have introduced multimodal conditions that combine textual and visual cues to guide the inpainting process [13, 18, 28]. The primary objective of this task is to achieve natural foreground–background fusion while modifying subject attributes or actions and preserving identity. This presents a non-trivial challenge, as multimodal approaches often struggle with modality alignment, requiring the model to maintain subject identity consistency while preserving editability under textual guidance.

Recent studies in text-subject-guided image inpainting leverage attention mechanisms to blend text and visual cues [13], but reliance on attention mechanisms often weakens textual semantics, causing prompt-content mismatches. LAR-Gen [18] adopts a coarse-to-fine pipeline with RefineNet, while DreamMix [28] enhances text-guided editing via diverse text-image training. Yet, both methods are constrained by training domains and struggle to preserve identity attributes during structural edits. We define this issue as Faithfulness Hallucination, where reconstructed content misaligns with prompts or subjects, as shown in Fig. 1. This arises from the inherent gap between text and images—text encodes high-level semantics, while images retain fine-grained details, making cross-modal alignment prone to distortion. Moreover, most models depend on costly training or fine-tuning, resulting in inefficiency and loss of prior knowledge. Balancing quality and efficiency remains a persistent challenge.

To mitigate Faithfulness Hallucinations in text-subject-guided image inpainting and circumvent the limitations of data-intensive training and few-shot fine-tuning, we propose a zero-shot framework that incorporates a training-free image inpainting method for precise text-guided subject editing. Inspired by sketch-based structural guidance in traditional image inpainting [29, 12, 30, 15], our two-stage method uses sketches as intermediaries. The sketch is a simplified hand-drawn depiction that can offer an intuitive representation of geometric features through minimalistic lines. We use sketches as a bridge for cross-modal alignment, converting textual prompts into explicit structural contours that guide generation. By integrating sketches with the subject's original visual features, we ensure identity consistency, where sketches provide structural constraints, while details such as texture and color retain local authenticity. This approach maintains structural accuracy and subject identity while precisely aligning textual descriptions with visual content, effectively mitigating Faithfulness Hallucination arising from modality misalignment.

Our framework consists of two stages: (1) Text-Guided Sketch Generation, where a fine-tuned Stable Diffusion [20] model converts text into sketches; and (2) Sketch-Mediated Subject Inpainting, a training-free, attention-based process inspired by TF-ICON [17]. In this stage, both main and reference images are inverted into noise and combined for inpainting to preserve identity. Fused self-attention and cross-attention maps from background and sketch reconstructions guide early generation, while later refinement relies solely on text prompts to enhance fine details. The Sketch-Edge Attention Enhancement further sharpens structural control by amplifying edge-focused attention.

In summary, our key contributions are as follows:

1. We introduce a novel method that utilizes sketches as an intermediary for image inpainting, effectively reducing Faithfulness Hallucination caused by modality misalignment and offering precise control.
2. We propose a zero-shot image inpainting framework that enables efficient restoration with minimal training cost.
3. We present a Sketch-Edge Attention Enhancement Strategy, focusing on structural elements and avoiding interference with color and texture.
4. Experiments show that our method reduces training costs, achieves an $80\times$ faster response than few-shot approaches, and outperforms existing alternatives in subject identity preservation (+15.8%) and text-guided structural editing accuracy (+16.4%).

2 Related Work

Text-Subject-Guided image inpainting. Text-subject-guided image inpainting overcomes the limitations of single-modal methods that lack fine-grained control and editing flexibility. Some subject-guided approaches [21] incorporate text to reduce ambiguity but still lack active editing. In contrast, this task aims for natural foreground–background fusion while modifying subject attributes or actions and preserving identity. Though recent work [13] uses attention mechanisms for this purpose, it underutilizes text semantics. DreamMix [28] enhances

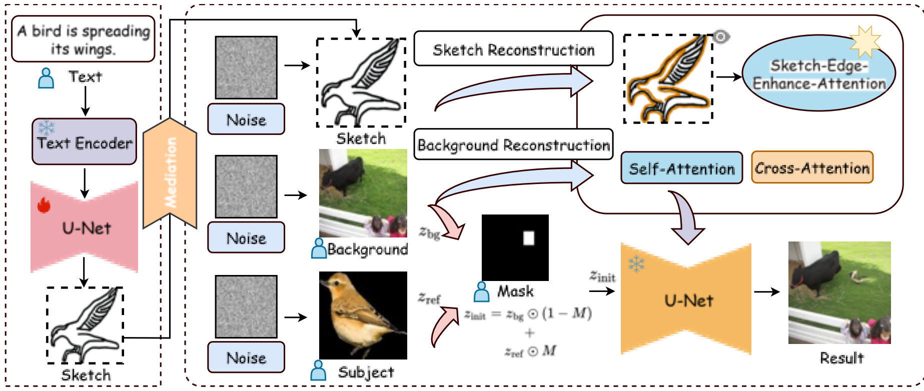

Fig. 2. Overall framework. Our framework includes two stages: the text-guided sketch generation (left) and the sketch-mediated subject-guided inpainting (right). A sketch-edge attention enhancement further refines structural focus. Flames and snowflakes denote learnable and frozen parameters, while the blue user icon indicates user input.

text-guided editing via learned text-image mappings, and LAR-Gen [18] refines generated subjects with a trained RefineNet. However, their effectiveness is constrained by training data distribution, often failing to balance identity preservation with semantic editing. Overall, existing methods suffer from Faithfulness Hallucination—a modality misalignment that disrupts consistency among generated content, subject identity, and textual intent.

Sketch-based image inpainting. Sketch-based image inpainting leverages hand-drawn sketches as structural guidance to restore missing regions. Unlike edge-based methods, sketches capture key geometry without strict pixel alignment, reducing user effort and improving controllability. Many works [29, 12, 30, 15] adopt sketches as auxiliary inputs. For example, DeepFill-v2 [29] employs gated convolutions, while SketchEdit [30] integrates sketch cues with style vectors for reconstruction. These approaches demonstrate the effectiveness of sketches in edge-level control. Building on this, we use sketches as a cross-modal bridge to exploit their structural expressiveness.

3 Methodology

We formalize the text-subject-guided image inpainting task as follows: Given an input quadruple $(I_{\text{ref}}, I_{\text{bg}}, M, t)$, where $I_{\text{ref}} \in \mathbb{R}^{H\times W\times 3}$ represents the reference subject image, $I_{\text{bg}} \in \mathbb{R}^{H\times W\times 3}$ is the background image to be inpainted, $M \in \{0,1\}^{H\times W}$ is a binary mask matrix identifying the missing region, and t is the textual instruction describing the subject object editing requirements. The task aims to satisfy the fundamental requirements of image inpainting [29, 8, 5], that is, to generate content that maintains visual coherence with the background, under the constraint $M \circ I_{\text{bg}}$, and to achieve twofold objectives: (1) The generated content must strictly adhere to the semantic editing requirements of I_{ref} as described by the text t; (2) The generated object must maintain consistency with the identity characteristics of I_{ref}.

Considering that sketches offer edge-level control, we innovatively treat them as a key medium for transforming text and visual information. As illustrated in Fig. 2, we propose a two-stage text-subject-guided image inpainting framework. In the first stage, a fine-tuned text-to-image model generates sketches from text, converting semantic cues into structural references. In the second stage, a training-free attention-based inpainting framework leverages sketches for precise structural control, aided by a sketch-edge attention enhancement that restricts edits to object boundaries without affecting color or texture.

3.1 Text-Guided Sketch Generation

Text-guided sketch synthesis remains challenging and follows two paradigms: raster-based generation through deep learning [24, 9, 10], which struggles with complex semantics, and vector-based generation [6, 11, 26, 25], which optimizes Bézier paths for high-precision design. However, vector approaches require per-image optimization, leading to low efficiency. In our setting, sketches function as structural references rather than precise design outputs.

Therefore, we propose a text-guided raster sketch generation approach that fine-tunes a pre-trained Stable Diffusion model to produce sketch-style outputs from texts. Leveraging the extensive visual–linguistic priors learned from billions of image–text pairs, our method achieves high-quality, robust, and generalizable sketch generation. Unlike vector-based approaches that rely on per-image Bézier optimization, our diffusion-based method enables single-pass text-to-sketch synthesis, improving generation efficiency.

Data Generation. Due to the absence of public text-scene sketch datasets, we created T2S-Sketch-8K, the first benchmark dataset for text-guided scene sketch synthesis. It includes 8k text-sketch pairs, addressing the lack of aligned sketch-text resources in multimodal generation research.

Fine-Tune. The framework employs a conditional 2D UNet to model latent-space image generation. Given the text prompt c, the CLIP encoder converts it into semantic embeddings, which are concatenated with noisy latent z_t as UNet inputs. The diffusion process follows:

$$z_t = \sqrt{\alpha_t} z_0 + \sqrt{1-\alpha_t}\epsilon, \quad \epsilon \sim \mathcal{N}(0, I)$$

where α_t is governed by the noise schedule. Training minimizes noise prediction error:

$$\mathcal{L} = \mathbb{E}_{z_0,\epsilon,t,c}\left[\|\epsilon - \epsilon_\theta(z_t, t, c)\|_2^2\right]$$

Fine-tuned on T2S-Sketch8k with a frozen text encoder, our UNet decoder uses cross-attention for text-sketch alignment. This module significantly improves generation efficiency while preserving text-structure alignment, producing clearer and more structured results without cluttered strokes (Fig. 3), offering crucial geometric guidance for inpainting.

3.2 Sketch-Mediated Subject Inpainting

In the second stage, we propose a training-free Sketch-Mediated Subject Inpainting Framework that employs sketches as structural constraints within a pre-trained diffusion model. Our sketch-edge attention strategy strengthens guidance on key edge regions, enhancing structural control and feature preservation.

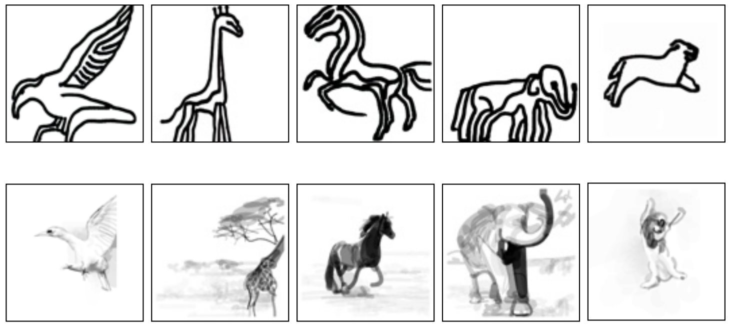

Fig. 3. Results of the sketch generation. Our method delivers clearer, more structured contours from text, outperforming vector-based approaches like DiffSketcher [25].

Image Inpainting Framework. Inspired by TF-ICON [17], our framework employs an "exceptional prompt" with zero semantics and a high-order ODE solver to invert the background and reference images into latent noise codes $\mathbf{z}_{\text{bg}}$ and $\mathbf{z}_{\text{ref}}$. These codes, following the diffusion prior, are then fused to form the initialization state $\mathbf{z}_{\text{init}}$ for synthesis.

To ensure multimodal consistency, our framework introduces a composite attention injection mechanism during the early diffusion stage ($t > \tau_A \cdot T$). This mechanism consists of: (i) a multi-scale self-attention map $\mathbf{A}_{\text{bg}}$ for background reconstruction to maintain scene continuity; (ii) an enhanced self-attention map $\mathbf{A}_{\text{sketch}}$ for sketch-guided reconstruction to constrain object shape; and (iii) a cross-modal attention map $\mathbf{A}_{\text{cross}}$ for deep semantic and spatial fusion. The mechanism integrates sketch guidance by replacing the background's self-attention map with the sketch's in the target region, which is marked by the user mask. In the edge transition area, a cross-attention map is introduced to seamlessly integrate edge information with the background, ensuring a smooth transition between the inpainted region and its surroundings. Meanwhile, the self-attention map of the non-masked areas remains unchanged, preserving the consistency of the background. The refined composite attention map is then injected into the pre-trained diffusion model to guide the final image generation process.

In the later diffusion stages ($t \leq \tau_A \cdot T$), the process relies solely on textual prompts to leverage the prior knowledge of the pre-trained text-to-image model. This stage refines fine-grained details such as lighting and alleviates dependence on sketch quality. The transition point τ_A (empirically set to 0.4) is tunable, allowing control over the balance between sketch and text guidance. Even with imperfect sketches, the model preserves semantic alignment with the textual input, effectively mitigating the influence of sketch inaccuracies.

Sketch-Edge Attention Enhancement Strategy. To reduce texture interference from sketch feature injection, we propose a Sketch-Edge Attention En-

Table 1. Quantitative results for conditional image inpainting across three tasks, with our results in bold. In the subject-guided task, semantic alignment is omitted; in the text-guided task, subject identity constraints are removed; and in the text+subject-guided task, FID is excluded due to the absence of ground-truth images.

Task	Dataset	Method	Semantic Alignment	Identity Consistency		Image Quality
			CLIP-T (↑)	CLIP-I (↑)	DINO (↑)	FID (↓)
Subject Guidance	COCO-Val	Paint-by-Example	–	78.06	83.72	7.82
		AnyDoor	–	79.13	84.09	6.67
		Ours	–	**80.16** (+1.3%)	**85.23** (+1.4%)	**5.42** (-18.7%)
Text Guidance	OPA	SD-Inpainting	23.87	–	–	18.93
		Blended-Diffusion	22.57	–	–	17.46
		Ours	**25.54** (+7.0%)	–	–	**14.02** (-19.7%)
Text+Subject Guidance	OPA	TIGIC	22.81	64.37	78.11	–
		LAR-Gen	21.13	64.46	75.81	–
		DreamMix	23.78	65.81	78.49	–
		Ours	**27.68** (+16.4%)	**78.27** (+18.9%)	**85.41** (+12.7%)	–

hancement Strategy. This mechanism enhances structural control by amplifying attention weights along sketch edges, while minimizing their influence on color and texture. Specifically, a binary sketch edge mask $M_{\text{edge}} \in \{0,1\}^{H\times W}$ is extracted, where 1 marks edge regions. After bilinear downsampling to match the feature map resolution, we obtain an edge index set

$$\mathcal{I}_{\text{edge}} = \{(i,j) \mid M_{\text{edge}}(i,j) = 1\}.$$

Given an attention feature map $A \in \mathbb{R}^{B\times N\times L}$, where B is batch size, N is number of heads, and $L = H \times W$, attention weights at edge positions are amplified by a factor α (set to 5 in our experiments):

$$A^{(b,n)}_{\text{enhanced}}(i,j) = \begin{cases} A^{(b,n)}(i,j)\cdot\alpha & \text{if } (i,j) \in \mathcal{I}_{\text{edge}} \\ A^{(b,n)}(i,j) & \text{otherwise} \end{cases}$$

This mechanism amplifies the attention weights in the edge regions, forcing the model to prioritize alignment with the structural features of the sketch during the generation process.

4 Experiments

4.1 Implementation Details

For the first stage, we fine-tuned the Stable Diffusion v1.5 model on the T2S-Sketch-8K dataset for 500 training steps using a single NVIDIA RTX 4090 GPU. For the second-stage image inpainting module: 20 sampling steps are used; the latent channel count $c = 4$; downsampling factor $f = 16$; one image is generated per inference. The guidance scale is set to 2.5 to balance conditional and unconditional outputs. Thresholds $\tau_A = 0.4$ and $\tau_B = 0.8$ are used for composite attention and background filtering, respectively. All input images are resized to 512×512, and the model operates at this resolution.

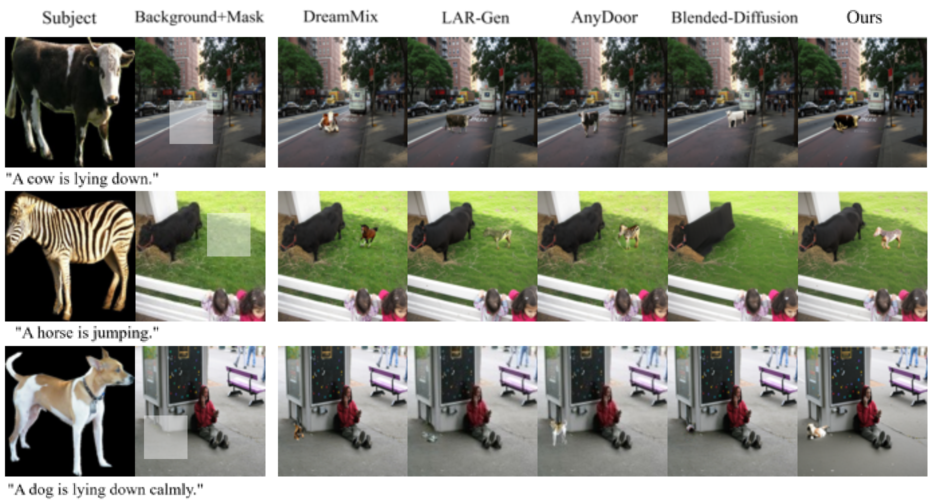

Fig. 4. Qualitative comparison including DreamMix [28], TIGIC [13], AnyDoor [4] and Blended-Diffusion [1], our method achieves superior performance in terms of text alignment accuracy and subject identity preservation.

4.2 Text-Subject-Guided Inpainting Evaluation

Test Benchmark. Due to the lack of publicly available text-subject-guided image inpainting test benchmarks, we built a dataset of 1,000 samples, each containing a background image and mask, a subject image and mask, and a textual description. Backgrounds were selected from OPA [16], with 20 subjects from 8 species and 5 editable text prompts per subject. Following DreamBooth [21], we evaluated identity consistency using CLIP-I [19] and DINO [3], and semantic alignment using CLIP-T [19]. Higher values across these metrics directly indicate reduced Faithfulness Hallucination, demonstrating stronger alignment between generated content and input conditions. Visual realism was further assessed via FID [7], where lower values denote more natural images.

Comparisons with Existing Alternatives. Our method achieves state-of-the-art performance in text-subject-guided image inpainting, outperforming existing approaches including DreamMix [28], LAR-Gen [18], and TIGIC [13], as evidenced by the quantitative metrics in Tab. 1. In particular, we intentionally selected subject-guided [4, 27] and text-guided [20, 1] methods as baselines, showing two extremes in unimodal image inpainting: high-fidelity identity preservation with subject images and open-ended structural editing with text semantics. This design effectively validates our method's robustness across different guidance scenarios, showing superior performance even in single-modal tasks.

Qualitative comparisons in Fig. 4 further reveal distinct advantages: While TIGIC and LAR-Gen struggle with text-driven structural editing, and DreamMix compromises subject identity to achieve semantic alignment, our method

Table 2. Comprehensive comparisons with existing alternatives. Results show our method achieves superior runtime efficiency and lower Faithfulness Hallucination Index.

Method	End-to-End Runtime(s) (↓)	Hallucination Index (↓)
TIGIC	21.8	7.03
LAR-Gen	15.5	8.15
DreamMix	1234.2	6.32
Ours	**14.1**	**1.95**

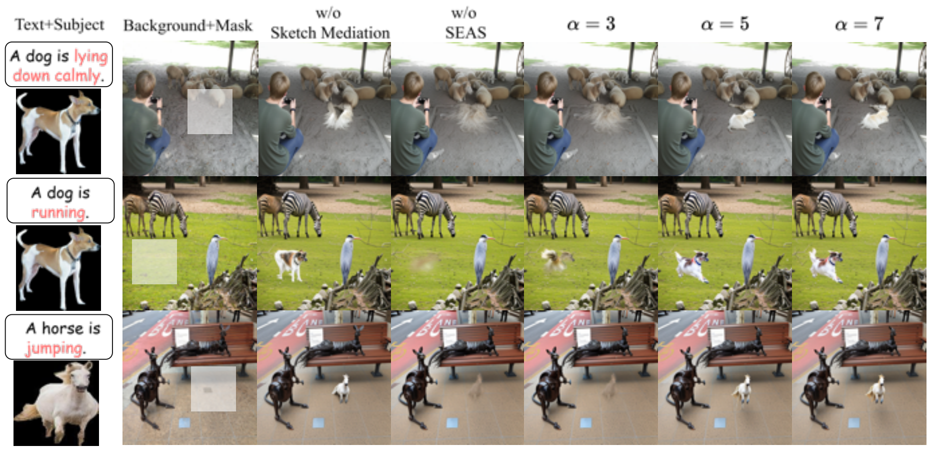

Fig. 5. Qualitative ablation studies on the core components of our method. "SEAS" denotes the Sketch-edge Attention Enhancement Strategy. The last three columns show the performance under different enhancement factors α.

uniquely preserves subject identity while precisely aligning with textual semantics, effectively reducing Faithfulness Hallucination.

Additionally, our comprehensive analysis (Tab. 2) highlights the superiority of our zero-shot method, which achieves faster end-to-end runtime and a lower Faithfulness Hallucination Index (HI). The proposed HI quantitatively measures hallucination by combining semantic alignment (CLIP-T) and identity consistency (CLIP-I, DINO) as:

$$\mathrm{HI} = 0.5 \times \text{CLIP-T}_{\mathrm{norm}} + 0.25 \times (\text{CLIP-I}_{\mathrm{norm}} + \mathrm{DINO}_{\mathrm{norm}})$$

ensuring a balanced evaluation of both efficiency and fidelity.

4.3 Ablation Study

We conducted ablation studies to verify the effectiveness of key components. Replacing the sketch attention map with that of the reference image removes sketch mediation, which preserves identity but limits semantic editing (Fig. 5).

Table 3. Quantitative results from ablation studies. The model achieves optimal performance when employing the sketch mediation combined with the sketch-edge attention enhancement strategy at the enhancement factor of 5, highlighted in gray.

Sketch Mediation	Sketch-Edge Attention Enhancement Strategy	CLIP-T (↑)	CLIP-I (↑)	DINO (↑)
✗	✗	23.61	73.35	82.98
✓	✗	24.93	72.35	82.07
✓	$\alpha = 3$	26.47	74.83	83.14
✓	$\alpha = 5$	27.68	78.27	84.80
✓	$\alpha = 7$	27.01	78.24	85.41

Table 4. Impact of Sketch Quality. Despite a 5.2% anomaly rate, the system shows minimal performance degradation, indicating strong robustness to imperfect sketches.

Sketch Status	Sample Size	CLIP-T (↑)	CLIP-I (↑)	DINO (↑)
Abnormal	52	26.31	71.93	80.84
Normal	948	27.75	78.61	85.66
Total	1000	27.68	78.27	85.41

As shown in Tab. 3, sketch mediation enhances text–image consistency and reduces Faithfulness Hallucination. The sketch-edge attention enhancement further improves performance, with the optimal factor at $\alpha = 5$; higher values cause distortion, while lower ones weaken edge control. This strategy effectively balances structural integrity and feature fidelity. The hyperparameter τ_A adjusts the trade-off between structural and semantic guidance, enabling generalization beyond purely structural edits.

4.4 Robustness Analysis

To assess the system's robustness to imperfect sketches, we measured the sketch anomaly rate (e.g., shape distortions, erratic strokes) with a threshold of $\tau_A = 0.4$. As shown in Tab. 4, the text prompt still offers semantic guidance despite sketch inaccuracies, reducing the negative impact of sketch errors on the final output. Notably, while text prompts aid image refinement, ablation results show that sketch-based guidance better preserves structural integrity.

5 Conclusion

Our zero-shot sketch-mediated framework aims to reduce Faithfulness Hallucination caused by modality misalignment and training costs in image inpainting. It first converts texts into structural sketches, bridging the gap between text and image modalities. A training-free refinement then leverages sketches and reference subjects with the sketch-edge attention enhancement strategy, ensuring subject identity consistency. As the first work to introduce sketch mediation, our framework improves text-visual alignment (+16.4%) and maintains subject identity (+15.8%) compared to SOTA, with minimal training overhead and an 80× faster response than few-shot methods.

References

1. Avrahami, O., Lischinski, D., Fried, O.: Blended diffusion for text-driven editing of natural images. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. pp. 18208–18218 (2022)
2. Bar-Tal, O., Ofri-Amar, D., Fridman, R., Kasten, Y., Dekel, T.: Text2live: Text-driven layered image and video editing. In: European conference on computer vision. pp. 707–723. Springer (2022)
3. Caron, M., Touvron, H., Misra, I., Jégou, H., Mairal, J., Bojanowski, P., Joulin, A.: Emerging properties in self-supervised vision transformers. In: Proceedings of the IEEE/CVF international conference on computer vision. pp. 9650–9660 (2021)
4. Chen, X., Huang, L., Liu, Y., Shen, Y., Zhao, D., Zhao, H.: Anydoor: Zero-shot object-level image customization. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. pp. 6593–6602 (2024)
5. Cong, W., Zhang, J., Niu, L., Liu, L., Ling, Z., Li, W., Zhang, L.: Dovenet: Deep image harmonization via domain verification. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. pp. 8394–8403 (2020)
6. Frans, K., Soros, L., Witkowski, O.: Clipdraw: Exploring text-to-drawing synthesis through language-image encoders. Advances in Neural Information Processing Systems **35**, 5207–5218 (2022)
7. Heusel, M., Ramsauer, H., Unterthiner, T., Nessler, B., Klambauer, G., Hochreiter, S.: Gans trained by a two time-scale update rule converge to a nash equilibrium. CoRR **abs/1706.08500** (2017), http://arxiv.org/abs/1706.08500
8. Hong, Y., Niu, L., Zhang, J.: Shadow generation for composite image in real-world scenes. In: Proceedings of the AAAI conference on artificial intelligence. vol. 36, pp. 914–922 (2022)
9. Huang, F., Canny, J.F.: Sketchforme: Composing sketched scenes from text descriptions for interactive applications. In: Proceedings of the 32nd annual ACM symposium on user interface software and technology. pp. 209–220 (2019)
10. Huang, F., Schoop, E., Ha, D., Canny, J.: Scones: towards conversational authoring of sketches. In: Proceedings of the 25th International Conference on Intelligent User Interfaces. pp. 313–323 (2020)
11. Jain, A., Xie, A., Abbeel, P.: Vectorfusion: Text-to-svg by abstracting pixel-based diffusion models. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. pp. 1911–1920 (2023)
12. Koley, S., Bhunia, A.K., Sekhri, D., Sain, A., Chowdhury, P.N., Xiang, T., Song, Y.Z.: It's all about your sketch: Democratising sketch control in diffusion models. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). pp. 7204–7214 (June 2024)
13. Li, P., Nie, Q., Chen, Y., Jiang, X., Wu, K., Lin, Y., Liu, Y., Peng, J., Wang, C., Zheng, F.: Tuning-free image customization with image and text guidance. In: European Conference on Computer Vision. pp. 233–250. Springer (2024)
14. Lim, S., Kim, S.: Image guided inpainting with parameter efficient learning. In: Proceedings of the IEEE/CVF International Conference on Computer Vision. pp. 1103–1111 (2023)
15. Liu, H., Wan, Z., Huang, W., Song, Y., Han, X., Liao, J., Jiang, B., Liu, W.: Deflocnet: Deep image editing via flexible low-level controls. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. pp. 10765–10774 (2021)

16. Liu, L., Liu, Z., Zhang, B., Li, J., Niu, L., Liu, Q., Zhang, L.: Opa: object placement assessment dataset. arXiv preprint arXiv:2107.01889 (2021)
17. Lu, S., Liu, Y., Kong, A.W.K.: Tf-icon: Diffusion-based training-free cross-domain image composition. In: Proceedings of the IEEE/CVF International Conference on Computer Vision. pp. 2294–2305 (2023)
18. Pan, Y., Mao, C., Jiang, Z., Han, Z., Zhang, J.: Locate, assign, refine: Taming customized image inpainting with text-subject guidance. arXiv e-prints pp. arXiv–2403 (2024)
19. Radford, A., Kim, J.W., Hallacy, C., Ramesh, A., Goh, G., Agarwal, S., Sastry, G., Askell, A., Mishkin, P., Clark, J., et al.: Learning transferable visual models from natural language supervision. In: International conference on machine learning. pp. 8748–8763. PmLR (2021)
20. Rombach, R., Blattmann, A., Lorenz, D., Esser, P., Ommer, B.: High-resolution image synthesis with latent diffusion models. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. pp. 10684–10695 (2022)
21. Ruiz, N., Li, Y., Jampani, V., Pritch, Y., Rubinstein, M., Aberman, K.: Dreambooth: Fine tuning text-to-image diffusion models for subject-driven generation. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. pp. 22500–22510 (2023)
22. Song, Y., Zhang, Z., Lin, Z., Cohen, S., Price, B., Zhang, J., Kim, S.Y., Aliaga, D.: Objectstitch: Object compositing with diffusion model. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. pp. 18310–18319 (2023)
23. Wang, S., Saharia, C., Montgomery, C., Pont-Tuset, J., Noy, S., Pellegrini, S., Onoe, Y., Laszlo, S., Fleet, D.J., Soricut, R., et al.: Imagen editor and editbench: Advancing and evaluating text-guided image inpainting. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. pp. 18359–18369 (2023)
24. Wang, Y., Chang, L., Cheng, Y., Jin, L., Cheng, Z., Deng, X., Duan, F.: Text2sketch: Learning face sketch from facial attribute text. In: 2018 25th IEEE International Conference on Image Processing (ICIP). pp. 669–673. IEEE (2018)
25. Xing, X., Wang, C., Zhou, H., Zhang, J., Yu, Q., Xu, D.: Diffsketcher: Text guided vector sketch synthesis through latent diffusion models. Advances in Neural Information Processing Systems **36**, 15869–15889 (2023)
26. Xing, X., Zhou, H., Wang, C., Zhang, J., Xu, D., Yu, Q.: Svgdreamer: Text guided svg generation with diffusion model. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. pp. 4546–4555 (2024)
27. Yang, B., Gu, S., Zhang, B., Zhang, T., Chen, X., Sun, X., Chen, D., Wen, F.: Paint by example: Exemplar-based image editing with diffusion models. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. pp. 18381–18391 (2023)
28. Yang, Y., Li, P., Zhang, L., Ma, L., Hu, P., Du, S., Zhuge, Y., Jia, X., Lu, H.: Dreammix: Decoupling object attributes for enhanced editability in customized image inpainting. arXiv preprint arXiv:2411.17223 (2024)
29. Yu, J., Lin, Z., Yang, J., Shen, X., Lu, X., Huang, T.S.: Free-form image inpainting with gated convolution. In: Proceedings of the IEEE/CVF international conference on computer vision. pp. 4471–4480 (2019)
30. Zeng, Y., Lin, Z., Patel, V.M.: Sketchedit: Mask-free local image manipulation with partial sketches. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. pp. 5951–5961 (2022)